Applied Machine Learning in Physics PHYS 2550
Homework 2 (Thu. Oct. 1, 2025) Due: Thu. Oct. 15, 2026

This assignment is about **matching the architecture to the data**. You will train a fully connected neural network (FCNN) and a two-dimensional convolutional neural network (2D CNN) on the *same* set of galaxy images, with the *same* preprocessing and the *same* training protocol, and investigate which properties of the data favour one architecture over the other. Much of the work, and many of the points, are in understanding the data before any network sees it: what the images contain, how noisy they are, and what the preprocessing does to them.

The assignment has four parts: theory (Part 1), data and preprocessing (Part 2), a controlled FCNN vs. CNN comparison (Part 3), and a short discussion (Part 4). Part 1 is closed-form reasoning of the kind you will meet in the midterm; answer it without running code.

Instructions

All coding should be done in Google Colab (preferred) or Jupyter notebooks, using PyTorch. Submit a **concise report in PDF format** on Canvas (not a dump of the code), with a link to the notebook used to produce every number and figure. Every plot must have labelled axes and a legend. Explanation and discussion of the results carry a large fraction of the points. Late submissions incur a penalty of 5% per calendar day (see the syllabus for late days). If AI tools made a material contribution to your work, add one or two sentences at the end saying which tools you used and for what.

Start early. Part 3 involves about 10 short training runs. Each is fast on a Colab GPU, but sessions time out and debugging is not compressible.

Dataset: Galaxy10 SDSS

The **Galaxy10 SDSS** dataset (distributed with the `astroNN` package) contains about 21 800 colour cutouts of galaxies, each 69×69 pixels in three SDSS bands (g, r, i), stored as 8-bit integers. Each image carries one of ten morphology labels derived from the volunteer votes of the Galaxy Zoo project:

Class	Morphology	Class	Morphology
0	Disk, face-on, no spiral	5	Disk, edge-on, boxy bulge
1	Smooth, completely round	6	Disk, edge-on, no bulge
2	Smooth, in-between round	7	Disk, face-on, tight spiral
3	Smooth, cigar shaped	8	Disk, face-on, medium spiral
4	Disk, edge-on, rounded bulge	9	Disk, face-on, loose spiral

Loading (either option):

- `pip install astroNN`, then
  ```
  from astroNN.datasets import load_galaxy10sdss
  images, labels = load_galaxy10sdss()
  ```
- or download `Galaxy10.h5` from the link in the astroNN documentation ("Galaxy10 SDSS") and read the datasets `images` and `ans` with `h5py`.

Save the arrays to your Google Drive after the first download so that you do not re-download them every session.

Reference architectures. Parts 1 and 3 use the following three models. All take the same preprocessed input: the central 48×48 pixels of each image (rows and columns 10–57), 3 channels.

- **CNN**: [Conv 3×3, 16 filters, padding 1, ReLU, MaxPool 2×2] $\rightarrow$ [Conv 3×3, 32 filters, padding 1, ReLU, MaxPool 2×2] $\rightarrow$ [Conv 3×3, 64 filters, padding 1, ReLU, MaxPool 2×2] $\rightarrow$ Flatten $\rightarrow$ Dense 64, ReLU, Dropout 0.3 $\rightarrow$ Dense 10.
- **FCNN-matched**: Flatten $\rightarrow$ Dense 24, ReLU $\rightarrow$ Dense 64, ReLU, Dropout 0.3 $\rightarrow$ Dense 10.

- **FCNN-large**: Flatten → Dense 256, ReLU → Dense 128, ReLU, Dropout 0.3 → Dense 10.

Part 1: Theory (30 points)

T1. Counting parameters. 3 Using $Z = XW + b$ for a dense layer and $[(k_h \times k_w \times C_{\text{in}}) + 1] \times C_{\text{out}}$ for a convolutional layer, compute the number of trainable parameters of each layer and the total for the three reference architectures. Which single layer holds most of the CNN's parameters, and what fraction of the total is it? Why does the FCNN-matched model need a first hidden layer as narrow as 24 nodes to stay within the CNN's budget?

T2. Convolution and pooling by hand. 5 Consider the single-channel input I and the kernel K (the "sharpening" kernel from lecture 4):

$$I = \begin{pmatrix} 1 & 2 & 0 & 1 & 3 \\ 0 & 3 & 1 & 2 & 1 \\ 2 & 1 & 4 & 1 & 0 \\ 1 & 0 & 2 & 3 & 1 \\ 0 & 1 & 1 & 2 & 2 \end{pmatrix}, \qquad K = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{pmatrix}.$$

(a) Compute the feature map for stride 1 and no padding. What is its size?

(b) What is the size of the feature map with zero padding of 1? With stride 2 and no padding? Give the general formula.

(c) Apply 2×2 max pooling and 2×2 average pooling, both with stride 1, to the result of (a). Comment on how the two pooled maps would change if the bright central pixel of I moved by one pixel.

(d) The entries of K sum to 1. What does K do to a region of constant intensity, such as a flat sky background? By what factor does it multiply the standard deviation of uncorrelated pixel noise? What does this imply for sharpening images of faint galaxies?

(e) Deep-learning libraries implement cross-correlation (no kernel flip) and call it convolution. Why does that make no difference for K, and why does it make no difference in general once the kernel is learned?

T3. Shapes and receptive field. 4

(a) Give the shape (height × width × channels) of the output of every layer of the reference CNN for a $48 \times 48 \times 3$ input.

(b) The *receptive field* of a unit is the region of the input image that can influence its value. What is the receptive field (in input pixels) of one unit after the third convolution? A typical galaxy in this dataset spans 20–40 pixels. Does any unit in the convolutional stack "see" the whole galaxy? Which part of the network combines information across the whole image?

(c) If the first convolution instead used stride 2 and no padding, what would its output size be?

T4. Equivariance and invariance. 4

(a) Define translation *equivariance* and translation *invariance*. Show that a convolutional layer is translation equivariant (ignore image boundaries).

(b) Which elements of the reference CNN give it (approximate) invariance to small shifts? Why is the invariance only approximate?

(c) Why does an FCNN have neither property? Predict, qualitatively, how the accuracy of each model changes when the test images are shifted by a few pixels relative to the training images. (Part 3.2 tests this.)

T5. Scrambled pixels. 3 Draw one random permutation of the 48×48 pixel positions and apply this same permutation to every image, in training and in test. Both models are retrained from scratch on the scrambled images. Predict what happens to the accuracy of the FCNN and of the CNN, and explain why. What does this experiment say about the assumptions (the inductive bias) built into each architecture?

T6. Parameters are not compute. 4

(a) Estimate the number of multiply–accumulate operations (MACs) per image for the forward pass of the CNN and of FCNN-matched. (For a convolutional layer: output height $\times$ output width $\times$ $C_{\text{out}} \times k^2 C_{\text{in}}$.)

(b) The two models have almost the same number of parameters. Which one is more expensive per training step, and by roughly what factor? Explain the difference in terms of weight sharing.

(c) Given (b), in what precise sense can a CNN still "train faster" than an FCNN? State what you would measure to support that claim.

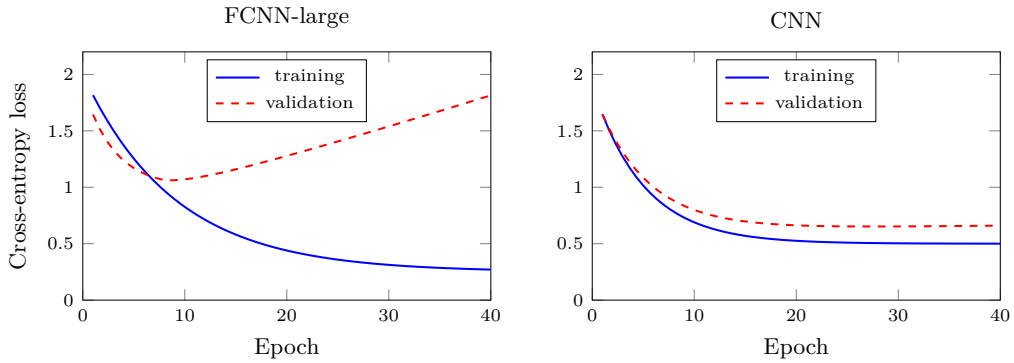
T7. Pooling and noise. 3 Consider a 2×2 pooling window containing only sky background: four independent pixel values with mean μ and standard deviation σ.

(a) What are the mean and standard deviation of the output of average pooling?

(b) Is the output of max pooling biased? In which direction? (For Gaussian noise, $\mathbb{E}[\max] \approx \mu + 1.03\,\sigma$.)

(c) Lecture 4 asked which pooling you would use for a dark object on a light background. For faint galaxies on a noisy sky, argue for one choice of pooling in the *first* layer.

T8. Reading training curves. 4 The figure shows the training and validation loss of FCNN-large (left) and of the CNN (right) on a galaxy classification task.



(a) Diagnose each model (under-fitting, over-fitting, or neither). At which epoch would early stopping select the FCNN?

(b) Name two changes that would reduce the gap for the FCNN, and say what each one does.

(c) Why is the gap between training and validation loss much smaller for the CNN, even though it trains on exactly the same images?

Part 2: Data and preprocessing (25 points)

Treat this part as an investigation of the dataset: every preprocessing choice you make must be justified by something you measured here.

2.1 First look. 5 Load the data and report the array shapes, the data type and the range of the pixel values. Show a grid with a few example images of every class. Tabulate the number of images per class.

Which classes are rare? Decide what to do with the rarest class (keep it, or drop it and work with nine classes), state your choice, and use it consistently. Make a stratified split into training (70%), validation (15%) and test (15%) sets with a fixed random seed. The test set is not to be looked at again until the final evaluations of Part 3.

2.2 Is the dataset noisy? 10

(a) *Sky noise.* For each image and each band, use the pixels in the outer 4-pixel frame of the 69×69 cutout to estimate the background level (median) and the background noise σ_{sky}. Estimate σ_{sky} both with the standard deviation and with the robust estimator $1.4826 \times \mathrm{MAD}$ (median absolute deviation). Plot the distributions. Why do the two estimators disagree for some images?

(b) *Signal-to-noise.* Define a simple signal-to-noise ratio per image: the background-subtracted flux inside a central aperture of radius 8 pixels, divided by $\sigma_{\mathrm{sky}}\sqrt{N_{\mathrm{pix}}}$. Plot its distribution per class. Are some classes systematically fainter?

(c) *Contamination and digitisation.* Using your σ_{sky} estimates, what fraction of images has a bright neighbouring source (e.g. a star) in the border frame? The images are 8-bit: what fraction of pixels sits at the maximum value (255)? Show one example of each and say in two or three sentences what information these effects remove.

(d) *Label noise.* The labels come from volunteer votes, not from ground truth. Which pairs of classes do you expect volunteers to confuse most? What does label noise imply for the best accuracy any classifier can reach on this test set?

Summarise in a few sentences: *does this dataset have noise, of which kinds, and which of them can preprocessing address?*

2.3 Preprocessing pipeline. 7 Build one preprocessing function used by *both* models:

(a) *Crop.* Measure the average radial light profile of the galaxies and use it to justify a crop size. The reference architectures assume the central 48×48 pixels; you may choose a different size if you justify it, but then use it for both models.

(b) *Normalisation.* Convert to floating point and standardise each band using the mean and standard deviation computed *on the training set only.* Explain why statistics from the full dataset must not be used. Compare with per-image normalisation (e.g. dividing each image by its brightest pixel, as discussed in lecture 4): what physical information does per-image normalisation remove, and is that information useful for this task?

(c) *Background.* Decide whether to subtract the sky level estimated in 2.2(a). Justify your choice.

(d) *Class imbalance.* Choose between class-weighted cross-entropy, a weighted sampler, or neither. Explain why plain accuracy can be misleading here and which additional metric (e.g. balanced accuracy or macro-averaged F1) you will report.

Show a few images before and after preprocessing, and confirm the per-band mean and standard deviation of the processed training set.

2.4 Augmentation and the symmetries of the sky. 3 Lecture 4 listed geometric, colour-space and kernel-filter augmentations. For galaxy images, which transformations correspond to true symmetries of the problem (the label does not change), and which would change the physics? Consider at least rotations by arbitrary angles, rotations by multiples of 90°, horizontal and vertical flips, small shifts, zoom, changes of the relative intensity of the g, r, i bands, and blurring. Which ones would you use, and why do the flips remain valid even though they change the handedness of a spiral?

Part 3: FCNN vs. 2D CNN on equal footing (35 points)

A comparison is only meaningful if the two models differ *only* in their architecture. Use the following protocol for every run in this part, and state it in your report:

- identical training, validation and test sets and identical preprocessing (Part 2), with no augmentation unless stated;
- identical loss (with or without the class weights chosen in 2.3), optimiser (Adam, learning rate 10^{-3}), batch size (128), and maximum number of epochs (50);
- early stopping on the validation loss with patience 5, restoring the best weights;
- the same hardware for all timing measurements (state the GPU type). Call `torch.cuda.synchronize()` before reading the clock.

3.1 Baseline comparison. 15 Implement the three reference models and verify their parameter counts against your answer to T1 (e.g. with `sum(p.numel() for p in model.parameters())`). Train each with 3 different random seeds. Report, as mean ± standard deviation over seeds:

- test accuracy and balanced accuracy (or macro F1);
- number of parameters;
- wall-clock time per epoch, and the number of epochs and wall-clock time needed to first reach a common target validation accuracy (use the best validation accuracy reached by FCNN-matched as the target);
- the epoch selected by early stopping.

Show the training and validation loss curves, the confusion matrix of the best seed of each model, and the one-vs-rest ROC curves for two classes of your choice. Which classes are confused, and does this match your expectations from 2.2(d)?

3.2 Shifted galaxies. 10 Without retraining, evaluate the models of 3.1 on test images cropped with the 48×48 window displaced by $d = 0, 2, 4, 6, 8$ pixels (in a random direction for each image) from the centre of the 69×69 cutout. Plot test accuracy against d for the three models and compare with your prediction in T4.

3.3 Scrambled pixels. 10 Carry out the experiment of T5: apply one fixed random pixel permutation to all images, retrain the CNN and FCNN-matched, and compare their test accuracy with the unscrambled baseline. Show one scrambled image. Was your prediction right?

Optional extensions (up to +5 bonus points)

Choose one.

B1. How much data does each model need? Retrain the CNN and FCNN-matched on 10%, 25% and 50% of the training set (stratified subsets; fixed validation and test sets). Plot test balanced accuracy against training-set size on a logarithmic axis and relate the result to weight sharing.

B2. Robustness to noise. Add Gaussian noise with standard deviation $k\,\sigma_{\mathrm{sky}}$, $k = 0, 1, 2, 4$, to the raw test images (before normalisation; clip to the valid pixel range) and evaluate the models of 3.1 without retraining. Plot accuracy against k and relate your answer to T7.

B3. Augmentation. Retrain the CNN and FCNN-matched with the augmentations you selected in 2.4 (random shifts, or flips and 90° rotations). Compare accuracy, the training/validation gap, and (for shifts) the curve of 3.2. Which model gains more, and why?

Part 4: Discussion (10 points)

In at most one page:

(a) Summarise Part 3 in one table (models as rows; accuracy, balanced accuracy, parameters, MACs, time per epoch, time to target accuracy as columns). Which model performs better? Is either one "faster", and in which precise sense?

(b) Which properties of the data (locality, translation symmetry, hierarchical structure, noise) favour one architecture over the other for this dataset? Support each claim with a specific result from Parts 2–3.

(c) Suppose instead that each SDSS object were described only by its five magnitudes u, g, r, i, z (a fixed-length feature vector). Would a 2D CNN be a sensible choice for classifying such data? Would a 1D CNN over the five bands? Justify with the same arguments.

(d) Given the noise sources identified in 2.2, what limits the accuracy that remains? What would you change first to improve it: the architecture, the preprocessing, or the labels?